

# HAT-P-20b

R-1223

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-20\_250109 · created 2026-07-31T09:03:35+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460684.9184 +/- 0.0016 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.14 +/- 0.016                  |
| Transit depth (Rp/Rs)^2                           | 1.95 +/- 0.45 [%]               |
| Orbital Inclination (inc)                         | 85.98 +/- 0.31                  |
| Airmass coefficient 1 (a1)                        | 0.9821 +/- 0.0067               |
| Airmass coefficient 2 (a2)                        | 0.0133 +/- 0.0046               |
| Scatter in the residuals of the lightcurve fit is | 0.68 %                          |
| Best Comparison Star                              | #1 - [271, 144]                 |
| Optimal Aperture                                  | 3.88                            |
| Optimal Annulus                                   | 18.09                           |
| Transit Duration (day)                            | 0.0673 +/- 0.0056               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.14 $\pm$ 0.016 vs 0.1284 (deviation 0.73 $\sigma$ ) |
| Duration fit vs published            | 0.0673 d vs 0.0775 d (deviation 1.82 $\sigma$ )       |
| Mid-time O-C                         | -4.9 min                                              |
| Depth SNR                            | 8.8                                                   |
| Residual scatter                     | 0.68 %                                                |

### Light curve

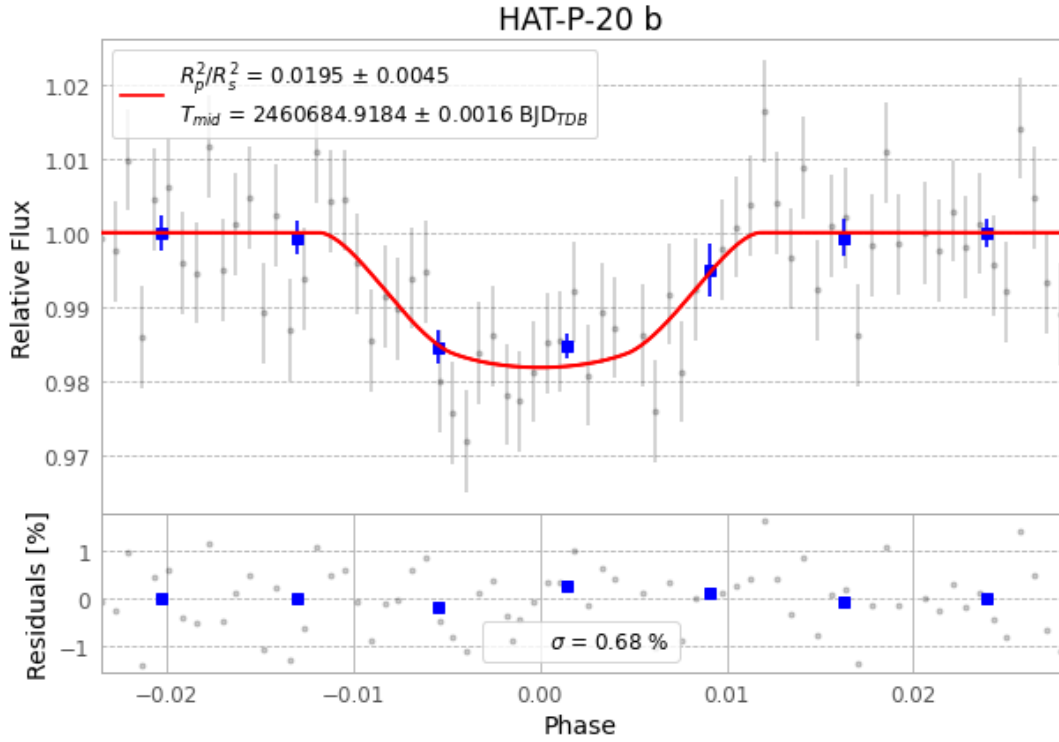


Figure 1 — FinalLightCurve\_HAT-P-20 b\_2025-01-09.png (SHA-256 ee72f6361c3ea85a...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [398, 277], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 e58391b0578a71b0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

71 FITS frames (47 MB), HATP-20250109081515.FITS → HATP-20250109114815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-250109.log (f81c54cb02f4...), FinalLightCurve\_HAT-P-20 b\_2025-01-09.png (ee72f6361c3e...), FinalLightCurve\_HAT-P-20 b\_2025-01-09.csv (c9f38ece8795...)

## Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 09:03Z.